



AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB)

Faculty of Engineering

BAE 2101: Computer Aided Design and Drafting

1.0 Title

Design a civil plan with electrical fitting & fixture according to BNBC of a Hospital Floor using AutoCAD

2.0 Objectives

In this experiment, student will design the civil plan of a **8000 square feet Hospital Floor** in AutoCAD with Electrical Fittings according to Bangladesh National Building Code.

3.0 Experiment to Solve

The experiment should be designed by the students following below mentioned requirements:

1.	Draw a Civil Plan of a Hospital Floor (total area approx. 8000 SFT) using AutoCAD Software. Doctor's room, Patient Waiting room, Appointment Area, Pharmacy, Staff dining area, Prayer room, toilets (Male and Female) must be included with other options. Proper dimensions should be maintained.	[P1]	20 points
2.	Draw the Electrical Fittings for the civil plan according to BNBC. Include electrical fixtures (switches, sockets, lights, fans, AC etc. as needed).	[P4] [P5]	10 points

4.0 Lab Project Submission

Course Name:	COMPUTER AIDED DESIGN & DRAFTING	Course Code:	BAE 2101
Semester:	Summer 2025-26	Section:	K
Faculty:	Raja Rashidul Hasan	OEL/Project	1
Assessment:	OEL (CO3), POI: P.e.1.C6		
Submission Deadline:	27 th August 2026		

OEL Exam Evaluation Rubrics:

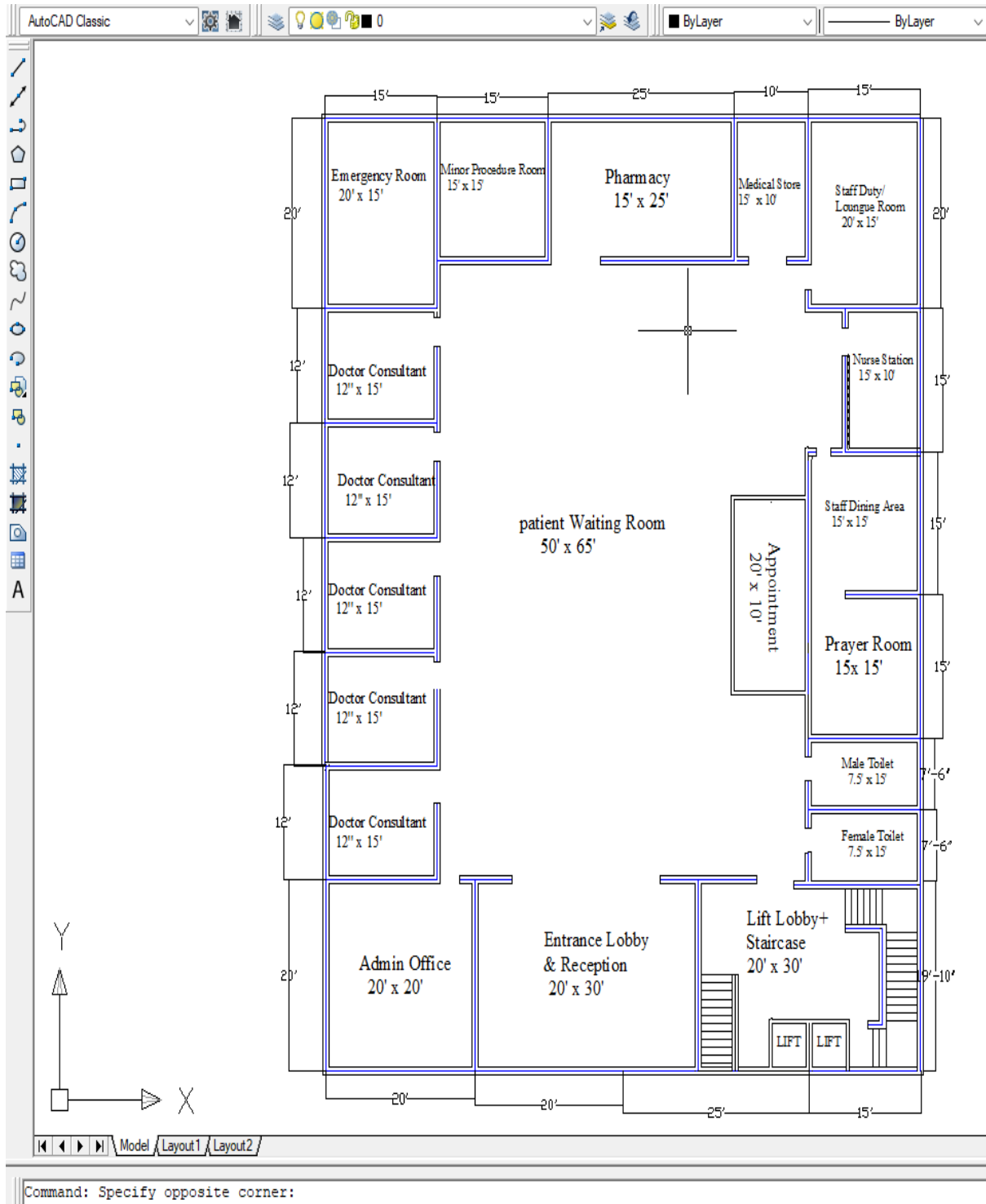
Category	Unsatisfactory (1-5)	Below Average (6-10)	Good (11-15)	Excellent (16-20)	Secured Marks
Civil Plan	The civil plan is either copied or very poor with major errors.	The civil plan is below average or incomplete with major errors.	The civil plan is drawn partially as per requirement with minor errors	The civil plan is unique and drawn as per requirements with proper dimensions	
Comments	Total Marks:				

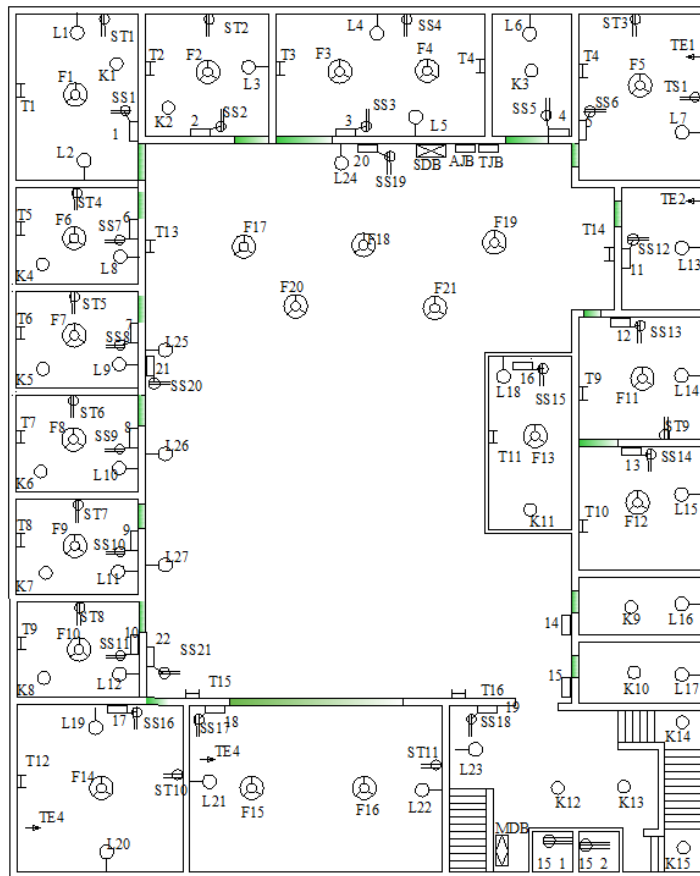
Category	Unsatisfactory (0-3)	Below Average (5)	Good (7)	Excellent (8-10)	Secured Marks
Electric Fittings	The fittings are incomplete or copied	The fittings are drawn or placed inappropriately without maintaining BNBC	The fittings are drawn & placed appropriately with partially complying to BNBC	The fittings are placed appropriately and maintaining BNBC	
Comments		Total Marks:			

OEL Viva Evaluation Rubrics

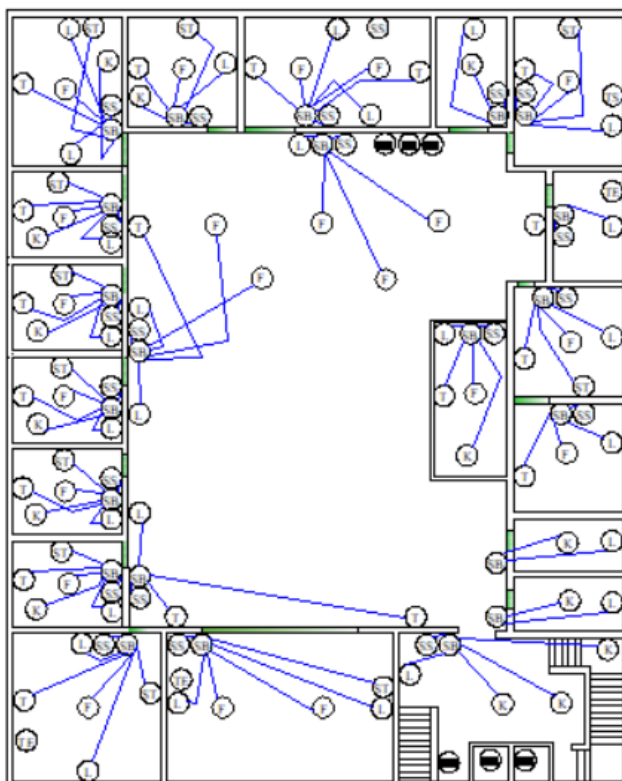
Category	Performance Levels	Marks
1. Conceptual Understanding (Civil Plan)	0: No relevant answer or cannot identify basic layout features. 1: Very poor understanding; can't explain room functions/arrangement. 2: Limited idea; can mention room names but no explanation of design purpose. 3: Basic understanding; can describe some parts/misses their justification. 4: Explains most spaces correctly with minor confusion. 5: Explains layout clearly with purpose; demonstrates full understanding of civil plan.	0–5
2. Understanding of Electrical Layout	0: Cannot explain or identify any electrical fittings. 1: Major misunderstanding or random placement. 2: Mentions few fittings without logic or reasoning. 3: Basic understanding of fittings but limited reasoning/ missing items. 4: Mostly correct placement and reasoning; minor errors in justification. 5: Correctly explains all electrical fittings with logical reasoning and clear relation to room usage; includes safety considerations.	0–5
3. Relation Between Civil and Electrical Plan	0: No understanding of relationship between the two plans. 1: Weak connection or incorrect explanation. 2: Limited connection shown; partial understanding. 3: Explains relation in most areas but misses a few key points. 4: Clearly justifies how electrical layout aligns with civil plan; demonstrates proper coordination.	0–4
4. Clarity & Communication Skill	0: Unable to express or explain ideas clearly. 1: Hesitant or unclear explanation; poor vocabulary. 2: Communicates fairly well with minor misuse of terms. 3: Explains fluently with correct technical terms; organized answers.	0–3
5. Response to Questions & Practical Insight	0: Fails to answer or gives irrelevant responses. 1: Answers few questions or guesses without clear logic. 2: Answers most questions correctly; minor gaps. 3: Responds accurately and confidently with practical reasoning.	0–3
Comments	Total Marks	

Serial	ID	Student Name	Department	Marks
22	23-54386-3	Tajmun Nahar Tisha	BSc CSE	
28	24-56318-1	MD. TANVER AHMED	BSc CSE	
9	23-52568-2	SADMAN ISLAM SHAMS	BSc CSE	
20	23-54308-3	SRABONI PAL	BSc CSE	

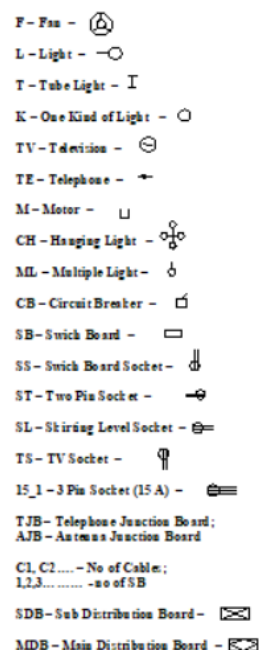
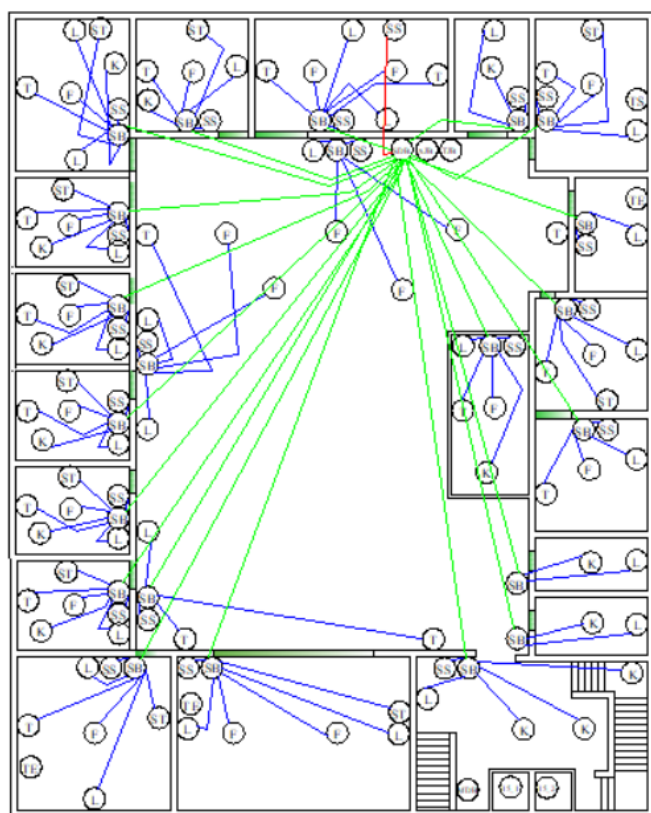
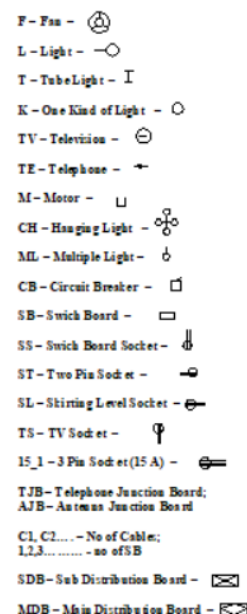
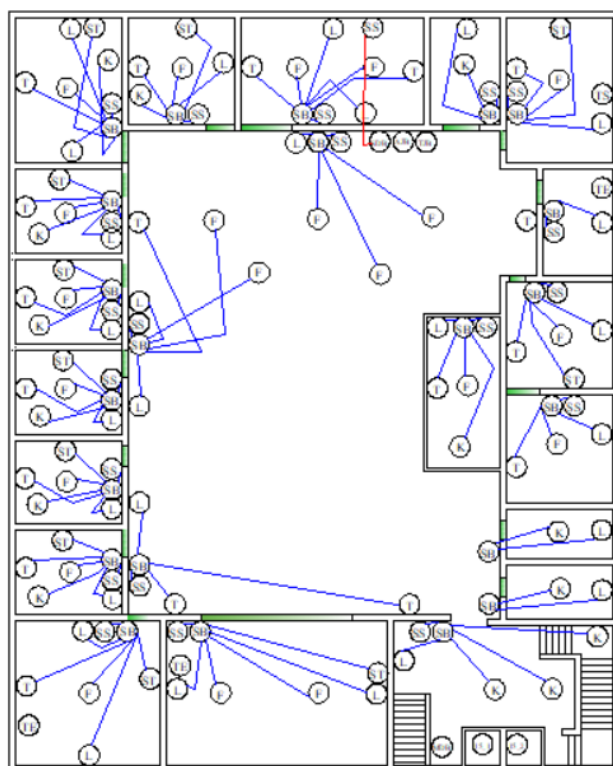


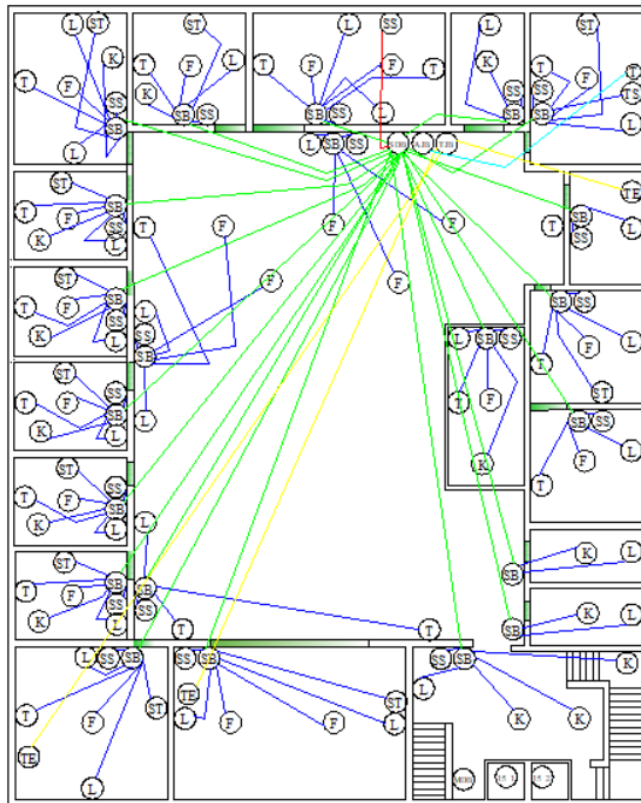


- F - Fan - 
- L - Light - 
- T - Tube Light - 
- K - One Kind of Light - 
- TV - Television - 
- TE - Telephone - 
- M - Motor - 
- CH - Hanging Light - 
- ML - Multiple Light - 
- CB - Circuit Breaker - 
- SB - Switch Board - 
- SS - Switch Board Socket - 
- ST - Two Pin Socket - 
- SL - Skirting Level Socket - 
- TS - TV Socket - 
- 15_1 - 3 Pin Socket (15 A) - 
- TJB - Telephone Junction Board;
AJB - Antenna Junction Board
- C1, C2..... - No of Cables;
1,2,3..... - no of SB
- SDB - Sub Distribution Board - 
- MDB - Main Distribution Board - 

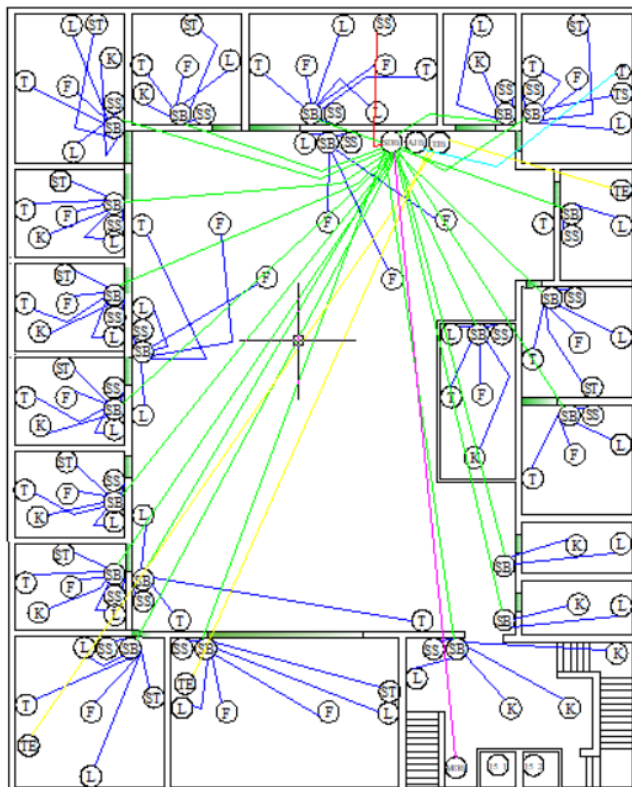


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